

WhatsApp AI Agent

An intelligent WhatsApp-based assistant designed to automate customer conversations, answer questions, qualify enquiries, capture leads, and hand over conversations to a human when required.

Prepared for Client / Business Name	Project Type AI + WhatsApp Automation
Platform WhatsApp Business	Status Proposed Project

Project Objective

Build a reliable AI agent that works through WhatsApp and handles routine customer conversations automatically while keeping the business team in control.

Confidential — prepared for client discussion

1. Project Overview

The proposed solution is a WhatsApp AI Agent that can communicate with customers in natural language, understand their questions, retrieve relevant business information, and take predefined actions. The goal is to reduce repetitive manual responses, improve response time, and make customer support available throughout the day.

The system will be designed around the client's business knowledge and workflows. The AI will not simply generate generic replies; it will be connected to approved business information and configured with clear rules for what it can answer, what actions it can perform, and when a conversation should be transferred to a human.

2. Proposed Solution

Component	Purpose
WhatsApp Integration	Receives and sends customer messages through the official WhatsApp Business/API integration.
AI Conversation Engine	Understands customer messages and generates context-aware responses.
Business Knowledge Base	Uses approved FAQs, services, policies, documents, and business information as the response source.
Lead Capture	Collects relevant customer details such as name, phone number, requirement, and enquiry information.
Human Handoff	Transfers conversations to a human when the AI cannot confidently handle the request or when escalation is required.
Admin / Monitoring	Provides a practical way to monitor conversations, update knowledge, and review AI activity.

3. Key Features

AI Customer Support Answers common customer questions in a conversational manner.	24/7 Automated Replies Handles routine enquiries without requiring a team member to respond manually.
Business-Specific Knowledge Grounds responses in client-provided information instead of relying only on general AI knowledge.	Lead Qualification Asks predefined questions and captures useful information from potential customers.
Service / Product Enquiries Provides information about approved services, products, pricing or other configurable details.	Conversation Context Maintains relevant context within a conversation so customers do not need to repeat information unnecessarily.
Human Escalation Routes sensitive, complex, or unsupported requests to a human representative.	Conversation Logging Stores relevant interaction data according to the agreed data-retention policy.
Admin Controls Allows approved business users to maintain knowledge and operational settings.	Analytics Tracks basic conversation, lead, and automation metrics where included in the final scope.

4. Example Customer Journey

STEP	FLOW	DESCRIPTION
01	Customer sends a WhatsApp message	Customer starts the conversation naturally.
02	AI identifies the intent	The agent determines whether the message is a FAQ, enquiry, lead, support request, etc.
03	AI retrieves relevant information	The system uses the configured business knowledge and rules.
04	AI responds / collects details	The customer receives an answer or is guided through the required questions.
05	Action or escalation	The system performs an approved action or transfers the conversation to a human.

5. Proposed Technology Architecture

- **WhatsApp Layer:** Official WhatsApp Business Platform / Cloud API or an approved provider.
- **Backend:** Node.js + Express.js for webhook handling, business logic, integrations, authentication, and APIs.
- **AI Layer:** LLM-based conversational engine with prompt/rule controls and business-context retrieval.
- **Knowledge Layer:** Structured business content and, where required, vector search / RAG for documents and FAQs.
- **Database:** MongoDB or PostgreSQL depending on the final data model and reporting requirements.
- **Frontend / Admin:** React-based dashboard if an administration interface is included in the final scope.
- **Deployment:** Cloud deployment with environment-based configuration, logging, and secure secret management.

6. Project Scope

AREA	INCLUDED WORK
Discovery & Planning	Understand business workflow, FAQs, customer journeys, escalation rules, and required integrations.
WhatsApp Setup	Configure webhook/API connectivity and message handling.
AI Agent Development	Develop the conversational agent, prompts, rules, context handling, and response flow.
Knowledge Integration	Connect approved business information and configure retrieval where required.
Lead & Workflow Automation	Implement agreed lead capture and business actions.
Human Handoff	Implement the agreed escalation mechanism.
Testing	Test common, edge-case, unsupported, and escalation scenarios.
Deployment	Deploy the application and complete production configuration.

7. Estimated Timeline

The timeline below is an initial estimate. The final schedule will depend on WhatsApp account/API approval, availability of business data, third-party integrations, and the final feature list.

PERIOD	PHASE	DELIVERABLES
Week 1	Discovery + architecture	Requirements, conversation flows, technical design
Week 2	WhatsApp integration	Webhook, message handling, basic end-to-end communication
Week 3	AI agent	Prompting, context, knowledge integration, core conversation flows
Week 4	Automation	Lead capture, workflows, escalation, integrations
Week 5	Dashboard + testing	Monitoring/admin features and comprehensive testing
Week 6	Deployment + handover	Production deployment, fixes, documentation, handover

8. Client Deliverables

- Configured WhatsApp AI Agent for the agreed business use cases.
- Backend/API and webhook implementation.
- Configured business knowledge / FAQ source.
- Agreed lead-capture and automation workflows.
- Human escalation flow.
- Production deployment.
- Basic technical and operational documentation.
- Handover and walkthrough for the client team.

9. Client Inputs Required

- WhatsApp Business account / required API access and verification information.
- Business profile, FAQs, product/service information, policies, and approved response content.
- Details of the target customers and common enquiry types.
- Lead fields and business actions the agent is allowed to perform.
- Human escalation contact/process.
- Access to any third-party systems that need to be integrated.

10. Commercial Proposal

ITEM	AMOUNT
AI Agent Development	To be finalized
WhatsApp / Meta / BSP Charges	Third-party charges — billed separately
AI Model / API Usage	Usage-based third-party charges — billed separately
Hosting / Infrastructure	Based on selected cloud infrastructure
Optional Maintenance & Support	To be finalized

11. Important Notes

- AI responses depend on the quality and completeness of the business information supplied for the knowledge base.
- WhatsApp, AI model, hosting, and other third-party services may have separate usage limits or charges.
- Exact features, integrations, SLA, support period, and pricing will be finalized after requirements confirmation.
- The agent will be configured to avoid taking actions outside the explicitly approved business workflow.
- Data retention, privacy, and access controls will be agreed with the client before production deployment.

Next Step

Confirm the business use cases, WhatsApp setup, required integrations, knowledge sources, and preferred customer journey. A final technical scope and commercial quotation can then be issued.